

Understanding Database Structure

Introduction

Before we start writing more SQL commands, it is important to understand how data is organized inside a database.

The main concepts we need to understand are:

- Database
- Table
- Row
- Record
- Column
- Field
- Data Types

Let us understand each concept using a simple Student Database example.

1. What is a Database?

Definition

A database is an organized collection of data that can be easily stored, accessed, managed, and updated.

In Simple Terms

Imagine a college has 1,000 students.

The college needs to store information about:

- Students
- Courses
- Teachers
- Marks
- Attendance

Instead of storing all this information randomly, the college stores it in an organized manner.

This organized collection of information is called a database.

Example

Suppose we create a database called:

```
college_database
```

Inside this database, we can store different types of information.

For example:

```
college_database  
  
    Students Information  
  
    Courses Information  
  
    Teachers Information  
  
    Marks Information
```

In an RDBMS, this information is usually organized into tables.

So our database may contain:

```
college_database  
  
    ↓  
  
students table  
  
courses table  
  
teachers table  
  
marks table
```

Simple Example

Think about a database as a cupboard.

The cupboard contains different shelves.

Similarly:

```
Cupboard = Database  
  
Shelves = Tables  
  
Items on the shelves = Data
```

Therefore, a database is the main storage place where related information is organized.

2. What is a Table?

Definition

A table is a collection of related data organized in the form of rows and columns.

In Simple Terms

A table looks similar to an Excel sheet.

For example, consider a Students table.

Student ID	Student Name	Age	Course
101	Sravya	25	Data Science
102	Ravi	24	Python
103	Priya	23	SQL

This is called a table.

The table contains information about students.

Table Name

The name of the above table can be:

students

Real-Life Example

Imagine a school attendance register.

The register contains information about many students.

The information is arranged in rows and columns.

Similarly, a database table stores information in rows and columns.

Simple Structure

DATABASE

↓

TABLE

↓

ROWS + COLUMNS

3. What is a Row?

Definition

A row represents one complete set of information about a single item or person in a table.

In Simple Terms

Look at the following table:

Student ID	Student Name	Age	Course
101	Sravya	25	Data Science
102	Ravi	24	Python
103	Priya	23	SQL

Consider this row:

101 | Sravya | 25 | Data Science

This row contains complete information about one student.

Therefore, it is called a row.

Another Example

102 | Ravi | 24 | Python

This is another row.

It contains complete information about Ravi.

Important Point

Each row usually represents one individual item.

For example:

In a Students table:

One Row = One Student

In an Employees table:

One Row = One Employee

In a Products table:

One Row = One Product

In an Orders table:

One Row = One Order

4. What is a Record?

Definition

A record is a complete collection of information about one particular item stored in a table.

In Simple Terms

A record and a row generally refer to the same concept in a database table.

Consider:

101 | Sravya | 25 | Data Science

This contains complete information about one student.

Therefore, it is called a record.

Row vs Record

Row = Record

Both terms are commonly used to describe one complete set of information in a table.

Example

Student ID	Student Name	Age	Course
101	Sravya	25	Data Science

The complete information:

101
Sravya
25
Data Science

together forms one record.

5. What is a Column?

Definition

A column represents a particular type or category of information stored in a table.

In Simple Terms

Consider the Students table:

Student ID	Student Name	Age	Course
101	Sravya	25	Data Science
102	Ravi	24	Python
103	Priya	23	SQL

The columns are:

Student ID
Student Name

Age

Course

Each column stores one particular type of information.

For example:

The Student ID column stores student IDs.

The Student Name column stores student names.

The Age column stores student ages.

The Course column stores course names.

Simple Example

Student ID → Stores IDs

Student Name → Stores Names

Age → Stores Ages

Course → Stores Course Names

6. What is a Field?

Definition

A field is a single piece of data stored at the intersection of a row and a column.

The term field is also sometimes used to refer to a column or attribute, depending on the context.

In Simple Terms

Consider the following table:

Student ID	Student Name	Age	Course
101	Sravya	25	Data Science

Here:

101

is one value.

Sravya

is one value.

25

is one value.

Data Science

is one value.

Each individual value can be called a field value.

Easy Way to Understand

Column = Type of Information

Field Value = Actual Information

For example:

Column Name → Student Name

Field Value → Sravya

Another example:

Column Name → Age

Field Value → 25

7. Difference Between Row and Column

Row	Column
Runs horizontally	Runs vertically
Contains complete information about one item	Contains one type of information
Also called a record	Also commonly called a field or attribute
One row may represent one student	One column may represent student names

Simple Example

Consider:

Student ID	Student Name	Age
101	Sravya	25
102	Ravi	24

The following is a row:

101 | Sravya | 25

The following is a column:

Student Name

Sravya

Ravi

Easy Memory Trick

ROW → Goes from Left to Right

COLUMN → Goes from Top to Bottom

8. What are Data Types?

Definition

A data type specifies the type of data that can be stored in a column.

In Simple Terms

Suppose we have the following information:

```
Student ID = 101  
  
Student Name = Sravya  
  
Age = 25  
  
Joining Date = 2026-02-25  
  
Course Fee = 25000.50
```

Look carefully.

All the values are not the same type.

```
101 → Whole Number  
  
Sravya → Text  
  
25 → Whole Number  
  
2026-02-25 → Date  
  
25000.50 → Decimal Number
```

The database needs to know what type of information each column will store.

For this purpose, we use Data Types.

9. Why Do We Need Data Types?

Imagine we create an Age column.

The Age column should contain values such as:

20

25

30

It should not normally contain:

Sravya

Data Science

Hello

Therefore, we specify that the Age column should store integer numbers.

Similarly:

Student Name → Text

Age → Integer

Joining Date → Date

Course Fee → Decimal Number

Data types help the database understand what kind of values should be stored in each column.

10. INT Data Type

Full Form

INT stands for Integer.

Definition

INT is used to store whole numbers.

In Simple Terms

Numbers without decimal points can be stored using INT.

Examples

```
1
25
100
5000
```

SQL Example

```
age INT
```

This tells the database:

"The age column should store whole numbers."

Another Example

```
student_id INT
```

The student_id column will store integer values.

For example:

```
101
102
103
```

Common Uses of INT

INT can be used for:

- Student ID
 - Employee ID
 - Age
 - Quantity
 - Number of Products
 - Marks
-

11. VARCHAR Data Type

Full Form

VARCHAR stands for Variable Character.

Definition

VARCHAR is used to store variable-length text or strings.

In Simple Terms

VARCHAR is used when we want to store text.

For example:

```
Sravya  
Data Science  
Hyderabad  
Python
```

SQL Example

```
student_name VARCHAR(50)
```

Here:

```
student_name
```

is the column name.

```
VARCHAR
```

is the data type.

```
50
```

is the maximum number of characters that can be stored.

Another Example

```
course VARCHAR(100)
```

This means that the Course column can store text up to 100 characters.

Common Uses of VARCHAR

VARCHAR can be used for:

- Names
 - Addresses
 - Course Names
 - City Names
 - Email Addresses
 - Product Names
-

12. DATE Data Type

Definition

DATE is used to store date values.

In Simple Terms

If we want to store a date, we can use the DATE data type.

Example

```
2026-02-25
```

The commonly used SQL date format is:

```
YYYY-MM-DD
```

Where:

```
YYYY → Year
```

```
MM → Month
```

DD → Day

SQL Example

```
joining_date DATE
```

This tells the database:

"The joining_date column should store date values."

Example Values

2026-02-25

2026-06-25

2025-12-10

Common Uses of DATE

DATE can be used for:

- Date of Birth
- Joining Date
- Order Date
- Delivery Date
- Appointment Date

13. DECIMAL Data Type

Definition

DECIMAL is used to store numbers that contain decimal points with a specified precision and scale.

In Simple Terms

DECIMAL is used when we need to store exact numerical values containing decimal points.

For example:

```
25000.50
```

99.99
450.75

SQL Example

```
course_fee DECIMAL(10,2)
```

Here:

10 → Total number of digits allowed
2 → Number of digits allowed after the decimal point

For example:

25000.50

contains:

5 digits before the decimal point
2 digits after the decimal point

Common Uses of DECIMAL

DECIMAL can be used for:

- Price
- Salary
- Course Fee
- Product Cost
- Financial Values

14. Understanding Data Types Using a Table

Consider the following Students table:

Student ID	Student Name	Age	Joining Date	Course Fee
101	Sravya	25	2026-02-25	25000.50
102	Ravi	24	2026-03-10	30000.00

The appropriate data types are:

Column	Data Type	Reason
Student ID	INT	Stores whole numbers
Student Name	VARCHAR(50)	Stores text
Age	INT	Stores whole numbers
Joining Date	DATE	Stores dates
Course Fee	DECIMAL(10,2)	Stores exact decimal numbers

15. Creating a Table Using These Data Types

Now let us combine everything we learned.

```
CREATE TABLE students
(
    student_id INT,
    student_name VARCHAR(50),
    age INT,
    joining_date DATE,
    course_fee DECIMAL(10,2)
);
```

Understanding the Code

Student ID

```
student_id INT
```

The student_id column stores whole numbers.

Student Name

```
student_name VARCHAR(50)
```

The student_name column stores text up to 50 characters.

Age

```
age INT
```

The age column stores whole numbers.

Joining Date

```
joining_date DATE
```

The joining_date column stores dates.

Course Fee

```
course_fee DECIMAL(10,2)
```

The course_fee column stores exact decimal values.

16. Understanding the Complete Database Structure

Let us connect everything we have learned.

DATABASE

↓

Contains TABLES

↓

TABLES contain ROWS and COLUMNS

↓

ROW represents one complete RECORD

↓

COLUMN represents one type of information

↓

FIELD VALUE represents one individual value

↓

DATA TYPE defines what kind of data a column can store

17. Complete Real-Life Example

Suppose we have:

college_database

Inside the database, we create:

students table

The table contains:

Student ID	Student Name	Age	Joining Date	Course Fee
101	Sravya	25	2026-02-25	25000.50
102	Ravi	24	2026-03-10	30000.00

Now let us identify each concept.

Database

college_database

Table

students

One Row or Record

101 | Sravya | 25 | 2026-02-25 | 25000.50

Columns

student_id

student_name

age
joining_date
course_fee

One Field Value

Sravya

Data Types

student_id → INT
student_name → VARCHAR(50)
age → INT
joining_date → DATE
course_fee → DECIMAL(10,2)

18. Quick Revision

Concept	Simple Meaning
Database	Organized collection of data
Table	Data arranged in rows and columns
Row	One complete set of information
Record	Another term for a complete row of information
Column	One particular category of information
Field Value	One individual value in a table
Data Type	Defines what kind of data a column can store
INT	Stores whole numbers
VARCHAR	Stores variable-length text
DATE	Stores dates
DECIMAL	Stores exact decimal numbers

19. Easy Memory Trick

Remember:

```
DATABASE
  ↓
TABLE
  ↓
ROWS + COLUMNS
  ↓
RECORDS + FIELD VALUES
  ↓
DATA TYPES
```

Or simply:

```
Database = Main Container

Table = Organized Sheet

Row = One Complete Record

Column = One Category

Field Value = One Individual Value

Data Type = Type of Value Allowed
```

20. Key Points to Remember

- A database is an organized collection of data.
 - A database can contain multiple tables.
 - A table contains rows and columns.
 - A row represents one complete record.
 - A column represents one category or attribute.
 - A field value is an individual value stored in a table.
 - Data types define what kind of data can be stored in a column.
 - INT is used for whole numbers.
 - VARCHAR is used for variable-length text.
 - DATE is used for dates.
 - DECIMAL is used for exact decimal values.
-

Conclusion

Understanding database structure is an important step before learning SQL commands.

The complete structure can be remembered as:

```
graph TD; Database --> Tables; Tables --> Rows_and_Columns[Rows and Columns]; Rows_and_Columns --> Records_and_Field_Values[Records and Field Values]; Records_and_Field_Values --> Data_Types[Data Types];
```

Database
↓
Tables
↓
Rows and Columns
↓
Records and Field Values
↓
Data Types

After understanding database structure, the next step is to learn how to create and manage databases using SQL commands such as:

- CREATE DATABASE
- SHOW DATABASES
- USE
- DROP DATABASE